

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q1 • Card 1/90

Which enzyme is responsible for the conjugation of bilirubin within the smooth endoplasmic reticulum of hepatocytes?

- A) UDP-glucuronosyltransferase (UGT1A1)
- B) Heme oxygenase
- C) Biliverdin reductase
- D) Glucose-6-phosphate dehydrogenase

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q1 • Card 1/90

Correct Answer

A

UDP-glucuronosyltransferase
(UGT1A1)

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q2 • Card 2/90

What is the immediate precursor to bilirubin in the heme breakdown pathway?

- A) Biliverdin
- B) Heme
- C) Urobilinogen
- D) Bilirubin diglucuronide

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q2 • Card 2/90

Correct Answer

A

Biliverdin

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q3 • Card 3/90

Where does unconjugated bilirubin primarily bind for transport in the bloodstream?

- A) Alpha-globulin
- B) Albumin
- C) Transferrin
- D) Haptoglobin

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q3 • Card 3/90

Correct Answer

B

Albumin

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q4 • Card 4/90

A newborn presents with jaundice on day 2 of life. Which physiologic mechanism is most likely responsible?

- A) Decreased UGT1A1 activity
- B) Increased heme oxygenase
- C) Biliary obstruction
- D) Genetic MRP2 defect

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q4 • Card 4/90

Correct Answer

A

Decreased UGT1A1 activity

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q5 • Card 5/90

Which form of bilirubin is water-soluble and can be excreted in urine?

- A) Unconjugated bilirubin
- B) Conjugated bilirubin
- C) Biliverdin
- D) Heme

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q5 • Card 5/90

Correct Answer

B

Conjugated bilirubin

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q6 • Card 6/90

Heme oxygenase cleaves heme to produce which two products?

- A) Biliverdin and carbon monoxide
- B) Bilirubin and iron
- C) Biliverdin and iron
- D) Bilirubin and carbon monoxide

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q6 • Card 6/90

Correct Answer

A

Biliverdin and carbon monoxide

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q7 • Card 7/90

Which enzyme converts biliverdin to bilirubin?

- A) Heme oxygenase
- B) UGT1A1
- C) Biliverdin reductase
- D) Glucuronidase

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q7 • Card 7/90

Correct Answer

C

Biliverdin reductase

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q8 • Card 8/90

What is the clinical significance of elevated unconjugated bilirubin in an adult?

- A) Possible hemolysis
- B) Possible biliary obstruction
- C) Possible hepatocellular carcinoma
- D) Possible Gilbert syndrome

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q8 • Card 8/90

Correct Answer

A

Possible hemolysis

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q9 • Card 9/90

Which of the following is NOT a cause of unconjugated hyperbilirubinemia?

- A) Hemolytic anemia
- B) Gilbert syndrome
- C) Crigler-Najjar syndrome
- D) Gallstone obstruction

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q9 • Card 9/90

Correct Answer

D

Gallstone obstruction

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q10 • Card 10/90

What is the normal range for total serum bilirubin in adults?

- A) 0.1 - 0.3 mg/dL
- B) 0.3 - 1.2 mg/dL
- C) 1.5 - 3.0 mg/dL
- D) 3.0 - 5.0 mg/dL

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q10 • Card 10/90

Correct Answer

B

0.3 - 1.2 mg/dL

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q11 • Card 11/90

**Which of the following statements
about UGT1A1 is TRUE?**

- A) It is located in the mitochondria
- B) It conjugates bilirubin with glucuronic acid
- C) It is activated by heme
- D) It is only expressed in newborns

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q11 • Card 11/90

Correct Answer

B

It conjugates bilirubin with
glucuronic acid

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q12 • Card 12/90

A patient with Gilbert syndrome will most likely have:

- A) Elevated conjugated bilirubin only
- B) Elevated unconjugated bilirubin with normal liver enzymes
- C) Elevated ALP and GGT
- D) Elevated ALT and AST

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q12 • Card 12/90

Correct Answer

B

Elevated unconjugated bilirubin
with normal liver enzymes

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q13 • Card 13/90

**The breakdown of heme from
which source contributes most to
bilirubin production?**

- A) Myoglobin
- B) Cytochrome P450
- C) Senescent red blood cells
- D) Catalase

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q13 • Card 13/90

Correct Answer

C

Senescent red blood cells

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q14 • Card 14/90

Bilirubin diglucuronide is formed by the action of UGT1A1 on which substrate?

- A) Biliverdin
- B) Unconjugated bilirubin
- C) Urobilinogen
- D) Heme

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q14 • Card 14/90

Correct Answer

B

Unconjugated bilirubin

EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)

Q15 • Card 15/90

Which condition is characterized by a complete absence of UGT1A1 activity?

- A) Gilbert syndrome
- B) Crigler-Najjar type I
- C) Dubin-Johnson syndrome
- D) Rotor syndrome

**EASY TIER (UGT1A1 Enzyme, Heme
Breakdown)**

Q15 • Card 15/90

Correct Answer

B

Crigler-Najjar type I

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q1 • Card 16/90

A 25-year-old man presents with mild jaundice during fasting. Labs show unconjugated bilirubin 3.5 mg/dL with normal liver enzymes. What is the most likely diagnosis?

- A) Viral hepatitis
- B) Gilbert syndrome
- C) Biliary obstruction
- D) Hemolytic anemia

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q1 • Card 16/90

Correct Answer

B

Gilbert syndrome

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q2 • Card 17/90

Which enzyme deficiency causes Crigler-Najjar syndrome type I?

- A) Heme oxygenase
- B) UGT1A1
- C) Biliverdin reductase
- D) Glucose-6-phosphate dehydrogenase

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q2 • Card 17/90

Correct Answer

B

UGT1A1

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q3 • Card 18/90

What is the fate of conjugated bilirubin in the intestine?

- A) Reabsorbed unchanged
- B) Converted to urobilinogen by gut bacteria
- C) Converted back to unconjugated bilirubin
- D) Excreted as biliverdin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q3 • Card 18/90

Correct Answer

B

Converted to urobilinogen by gut bacteria

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q4 • Card 19/90

A patient with Gilbert syndrome will show improvement in bilirubin levels with:

- A) Fasting
- B) Phenobarbital administration
- C) Alcohol consumption
- D) High-fat diet

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q4 • Card 19/90

Correct Answer

B

Phenobarbital administration

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q5 • Card 20/90

Which of the following bilirubin fractions is NOT measured directly by the van den Bergh reaction?

- A) Conjugated bilirubin
- B) Unconjugated bilirubin
- C) Delta bilirubin
- D) Direct bilirubin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q5 • Card 20/90

Correct Answer

B

Unconjugated bilirubin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q6 • Card 21/90

In hemolytic anemia, which bilirubin fraction is primarily elevated?

- A) Conjugated bilirubin
- B) Unconjugated bilirubin
- C) Direct bilirubin
- D) Bile acids

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q6 • Card 21/90

Correct Answer

B

Unconjugated bilirubin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q7 • Card 22/90

What is the rate-limiting step in bilirubin conjugation?

- A) Uptake of bilirubin into hepatocytes
- B) Activity of UGT1A1
- C) Transport into bile canaliculi
- D) Binding to albumin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q7 • Card 22/90

Correct Answer

B

Activity of UGT1A1

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q8 • Card 23/90

Which substance can induce UGT1A1 activity?

- A) Phenobarbital
- B) Rifampicin
- C) Alcohol
- D) Aspirin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q8 • Card 23/90

Correct Answer

A

Phenobarbital

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q9 • Card 24/90

A preterm infant with prolonged jaundice and kernicterus risk likely has:

- A) Physiologic jaundice
- B) Crigler-Najjar syndrome
- C) Biliary atresia
- D) Gilbert syndrome

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q9 • Card 24/90

Correct Answer

B

Crigler-Najjar syndrome

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q10 • Card 25/90

What is the primary route of elimination for unconjugated bilirubin?

- A) Urine
- B) Bile
- C) Feces
- D) Hepatic metabolism

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q10 • Card 25/90

Correct Answer

B

Bile

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q11 • Card 26/90

Bilirubin is toxic to which organ system at high concentrations?

- A) Cardiovascular
- B) Central nervous system
- C) Renal
- D) Pulmonary

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q11 • Card 26/90

Correct Answer

B

Central nervous system

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q12 • Card 27/90

Kernicterus is associated with deposition of bilirubin in which brain structure?

- A) Cerebellum
- B) Basal ganglia
- C) Cerebral cortex
- D) Brainstem

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q12 • Card 27/90

Correct Answer

B

Basal ganglia

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q13 • Card 28/90

Which of the following increases the risk of kernicterus in neonates?

- A) Albumin administration
- B) Phototherapy
- C) Acidosis
- D) Phenobarbital

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q13 • Card 28/90

Correct Answer

C

Acidosis

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q14 • Card 29/90

Phototherapy for neonatal jaundice converts bilirubin to:

- A) Biliverdin
- B) Lumirubin
- C) Urobilinogen
- D) Bilirubin diglucuronide

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q14 • Card 29/90

Correct Answer

B

Lumirubin

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q15 • Card 30/90

What is the half-life of bilirubin in the circulation?

- A) 1-2 hours
- B) 4-8 hours
- C) 12-24 hours
- D) 2-4 days

MID EASY TIER (Rapid-fire diagnostic interpretation of baseline conjugation)

Q15 • Card 30/90

Correct Answer

B

4-8 hours

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q1 • Card 31/90

Which bacterial enzyme converts bilirubin to urobilinogen in the intestine?

- A) Beta-glucuronidase
- B) Heme oxygenase
- C) UGT1A1
- D) Biliverdin reductase

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q1 • Card 31/90

Correct Answer

A

Beta-glucuronidase

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q2 • Card 32/90

What is the final oxidized product of urobilinogen in feces?

- A) Bilirubin
- B) Stercobilin
- C) Urobilin
- D) Biliverdin

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q2 • Card 32/90

Correct Answer

B

Stercobilin

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q3 • Card 33/90

**In complete biliary obstruction,
which of the following is TRUE?**

- A) Urinary urobilinogen is elevated
- B) Urinary urobilinogen is absent
- C) Fecal stercobilin is elevated
- D) Serum bilirubin is decreased

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q3 • Card 33/90

Correct Answer

B

Urinary urobilinogen is absent

MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)

Q4 • Card 34/90

Which condition leads to increased urinary urobilinogen?

- A) Biliary obstruction
- B) Hemolytic anemia
- C) Gilbert syndrome
- D) Dubin-Johnson syndrome

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q4 • Card 34/90

Correct Answer

B

Hemolytic anemia

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q5 • Card 35/90

The enterohepatic circulation of urobilinogen involves:

- A) Reabsorption from the gut and excretion in bile
- B) Conversion to bilirubin in the liver
- C) Direct excretion in urine
- D) Storage in the gallbladder

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q5 • Card 35/90

Correct Answer

A

Reabsorption from the gut and
excretion in bile

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q6 • Card 36/90

A patient with pale stools and dark urine likely has:

- A) Hemolytic anemia
- B) Biliary obstruction
- C) Gilbert syndrome
- D) Crigler-Najjar syndrome

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q6 • Card 36/90

Correct Answer

B

Biliary obstruction

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q7 • Card 37/90

Which of the following is the normal color of feces due to stercobilin?

- A) Pale yellow
- B) Brown
- C) Green
- D) Black

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q7 • Card 37/90

Correct Answer

B

Brown

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q8 • Card 38/90

In which condition would you expect elevated fecal stercobilin?

- A) Biliary atresia
- B) Hemolytic anemia
- C) Gallstone obstruction
- D) Cholangiocarcinoma

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q8 • Card 38/90

Correct Answer

B

Hemolytic anemia

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q9 • Card 39/90

Urinary urobilinogen is normally present in:

- A) Trace amounts
- B) Large amounts
- C) Absent completely
- D) Only in disease states

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q9 • Card 39/90

Correct Answer

A

Trace amounts

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q10 • Card 40/90

Which antibiotic use can affect urobilinogen production?

- A) Penicillin
- B) Ciprofloxacin
- C) Metronidazole
- D) Amoxicillin

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q10 • Card 40/90

Correct Answer

C

Metronidazole

MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)

Q11 • Card 41/90

Gut flora composition affects which metabolic process?

- A) Bilirubin synthesis
- B) Urobilinogen production
- C) Bilirubin conjugation
- D) Heme breakdown

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q11 • Card 41/90

Correct Answer

B

Urobilinogen production

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q12 • Card 42/90

What is the normal range for fecal stercobilin excretion?

- A) 50-250 mg/day
- B) 250-500 mg/day
- C) 500-1000 mg/day
- D) 1-5 g/day

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q12 • Card 42/90

Correct Answer

A

50-250 mg/day

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q13 • Card 43/90

**A patient with intrahepatic
cholestasis will have:**

- A) Increased urinary urobilinogen
- B) Decreased urinary urobilinogen
- C) Normal urinary urobilinogen
- D) Elevated fecal stercobilin

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q13 • Card 43/90

Correct Answer

B

Decreased urinary urobilinogen

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q14 • Card 44/90

Which of the following drugs can cause cholestasis and reduce urobilinogen?

- A) Phenobarbital
- B) Chlorpromazine
- C) Rifampicin
- D) Penicillin

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q14 • Card 44/90

Correct Answer

B

Chlorpromazine

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q15 • Card 45/90

What is the primary determinant of fecal color?

- A) Dietary intake
- B) Stercobilin concentration
- C) Bile acid concentration
- D) Bacterial overgrowth

**MEDIUM TIER (Urinary Urobilinogen,
Stercobilin)**

Q15 • Card 45/90

Correct Answer

B

Stercobilin concentration

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q1 • Card 46/90

**Which test best differentiates
biliary obstruction from
hepatocellular disease?**

- A) ALT/AST ratio
- B) ALP and GGT
- C) Serum bilirubin
- D) Prothrombin time

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q1 • Card 46/90

Correct Answer

B

ALP and GGT

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q2 • Card 47/90

**In a patient with jaundice, elevated
ALP with normal ALT suggests:**

- A) Hepatitis
- B) Biliary obstruction
- C) Hemolysis
- D) Gilbert syndrome

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q2 • Card 47/90

Correct Answer

B

Biliary obstruction

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q3 • Card 48/90

What is the hallmark of a physical block in the biliary tree?

- A) Elevated unconjugated bilirubin
- B) Elevated conjugated bilirubin with elevated ALP
- C) Normal ALP
- D) Elevated urobilinogen

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q3 • Card 48/90

Correct Answer

B

Elevated conjugated bilirubin with
elevated ALP

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q4 • Card 49/90

**A patient with choledocholithiasis
will have:**

- A) Elevated urobilinogen
- B) Decreased ALP
- C) Elevated ALP and GGT
- D) Normal bilirubin

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q4 • Card 49/90

Correct Answer

C

Elevated ALP and GGT

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q5 • Card 50/90

**Which of the following is a
metabolic flaw in bilirubin
handling?**

- A) Gallstone obstruction
- B) Gilbert syndrome
- C) Cholangiocarcinoma
- D) Biliary stricture

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q5 • Card 50/90

Correct Answer

B

Gilbert syndrome

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q6 • Card 51/90

In hepatocellular disease, which enzyme elevation pattern is most common?

- A) ALP > ALT
- B) ALT > ALP
- C) GGT > ALP
- D) AST > ALP

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q6 • Card 51/90

Correct Answer

B

ALT > ALP

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q7 • Card 52/90

Which of the following suggests a physical obstruction rather than a metabolic defect?

- A) Isolated unconjugated hyperbilirubinemia
- B) Conjugated hyperbilirubinemia with elevated ALP
- C) Elevated urobilinogen
- D) Normal liver enzymes

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q7 • Card 52/90

Correct Answer

B

Conjugated hyperbilirubinemia with
elevated ALP

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q8 • Card 53/90

In which condition would you see a normal fecal stercobilin despite jaundice?

- A) Biliary obstruction
- B) Hemolytic anemia
- C) Gilbert syndrome
- D) Crigler-Najjar syndrome

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q8 • Card 53/90

Correct Answer

B

Hemolytic anemia

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q9 • Card 54/90

What imaging modality is most useful to differentiate physical obstruction from metabolic flaw?

- A) CT scan
- B) Ultrasound
- C) MRI
- D) ERCP

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q9 • Card 54/90

Correct Answer

B

Ultrasound

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q10 • Card 55/90

**Dubin-Johnson syndrome
represents which type of defect?**

- A) Conjugation defect
- B) Transport defect
- C) Hemolytic defect
- D) Bacterial metabolism defect

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q10 • Card 55/90

Correct Answer

B

Transport defect

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q11 • Card 56/90

A patient with isolated elevated conjugated bilirubin with normal ALP likely has:

- A) Gilbert syndrome
- B) Dubin-Johnson syndrome
- C) Biliary obstruction
- D) Hemolytic anemia

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q11 • Card 56/90

Correct Answer

B

Dubin-Johnson syndrome

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q12 • Card 57/90

Which condition causes a physical block at the ampulla of Vater?

- A) Gilbert syndrome
- B) Pancreatic head tumor
- C) Crigler-Najjar syndrome
- D) Rotor syndrome

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q12 • Card 57/90

Correct Answer

B

Pancreatic head tumor

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q13 • Card 58/90

Which of the following is NOT a metabolic defect in bilirubin metabolism?

- A) Gilbert syndrome
- B) Crigler-Najjar syndrome
- C) Choledocholithiasis
- D) Dubin-Johnson syndrome

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q13 • Card 58/90

Correct Answer

C

Choledocholithiasis

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q14 • Card 59/90

**What is the characteristic finding
in Dubin-Johnson syndrome?**

- A) Black liver on biopsy
- B) White liver on biopsy
- C) Green liver on biopsy
- D) Normal liver on biopsy

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q14 • Card 59/90

Correct Answer

A

Black liver on biopsy

MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)

Q15 • Card 60/90

Which test result would you expect in a metabolic defect like Rotor syndrome?

- A) Elevated ALP
- B) Elevated GGT
- C) Isolated conjugated hyperbilirubinemia
- D) Elevated AST

**MID MEDIUM TIER (Speed differentiation of
physical blocks vs metabolic flaws)**

Q15 • Card 60/90

Correct Answer

C

Isolated conjugated
hyperbilirubinemia

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q1 • Card 61/90

Which enzyme ratio is most characteristic of hepatocellular injury?

- A) $ALT > ALP$
- B) $ALP > ALT$
- C) $GGT > ALP$
- D) $AST > ALT$

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q1 • Card 61/90

Correct Answer

A

ALT > ALP

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q2 • Card 62/90

In alcoholic liver disease, which ratio is typically elevated?

- A) $\text{ALT/AST} < 1$
- B) $\text{ALT/AST} > 1$
- C) $\text{ALP/GGT} < 1$
- D) $\text{GGT/ALP} > 1$

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q2 • Card 62/90

Correct Answer

A

ALT/AST < 1

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q3 • Card 63/90

Which of the following is the most sensitive marker for biliary obstruction?

- A) ALT
- B) AST
- C) GGT
- D) LDH

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q3 • Card 63/90

Correct Answer

C

GGT

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q4 • Card 64/90

A patient with gallstone obstruction will typically have:

- A) $ALT > AST$
- B) $ALP > ALT$
- C) $AST > ALT$
- D) GGT normal

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q4 • Card 64/90

Correct Answer

B

ALP > ALT

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q5 • Card 65/90

What is the AST/ALT ratio in viral hepatitis?

- A) < 1
- B) > 1
- C) 2:1
- D) 3:1

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q5 • Card 65/90

Correct Answer

A

< 1

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q6 • Card 66/90

Which enzyme is localized in the canalicular membrane of hepatocytes?

- A) ALT
- B) AST
- C) ALP
- D) LDH

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q6 • Card 66/90

Correct Answer

C

ALP

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q7 • Card 67/90

In alcoholic cirrhosis, which pattern is most common?

- A) $AST > ALT$ (ratio > 2)
- B) $ALT > AST$ (ratio > 2)
- C) $ALP > AST$
- D) $GGT > ALP$

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q7 • Card 67/90

Correct Answer

A

AST > ALT (ratio > 2)

Which of the following is NOT associated with elevated ALP?

- A) Biliary obstruction
- B) Bone disease
- C) Hepatocellular injury
- D) Pregnancy

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q8 • Card 68/90

Correct Answer

C

Hepatocellular injury

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q9 • Card 69/90

What is the half-life of ALP in serum?

- A) 1-2 days
- B) 3-7 days
- C) 10-14 days
- D) 20-30 days

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q9 • Card 69/90

Correct Answer

B

3-7 days

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q10 • Card 70/90

Which liver enzyme is most specific to the hepatobiliary system?

- A) ALT
- B) AST
- C) GGT
- D) LDH

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q10 • Card 70/90

Correct Answer

C

GGT

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q11 • Card 71/90

A patient with acute hepatitis presents with ALT 800 U/L and ALP 150 U/L. What is the most likely diagnosis?

- A) Biliary obstruction
- B) Viral hepatitis
- C) Alcoholic liver disease
- D) Gallstone disease

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q11 • Card 71/90

Correct Answer

B

Viral hepatitis

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q12 • Card 72/90

Which condition shows ALP elevation with normal GGT?

- A) Biliary obstruction
- B) Bone disease
- C) Hepatocellular injury
- D) Cholestasis

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q12 • Card 72/90

Correct Answer

B

Bone disease

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q13 • Card 73/90

In cholestatic liver disease, which enzyme elevation is more significant?

- A) ALT
- B) AST
- C) ALP
- D) LDH

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q13 • Card 73/90

Correct Answer

C

ALP

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q14 • Card 74/90

What is the R ratio used to differentiate hepatocellular from cholestatic injury?

- A) $(\text{ALT/ULN}) / (\text{ALP/ULN})$
- B) $(\text{ALP/ULN}) / (\text{ALT/ULN})$
- C) $(\text{AST/ULN}) / (\text{ALT/ULN})$
- D) $(\text{GGT/ULN}) / (\text{ALP/ULN})$

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q14 • Card 74/90

Correct Answer

A

$(ALT/ULN) / (ALP/ULN)$

Which of the following is TRUE about GGT?

- A) It is elevated in bone disease
- B) It is induced by alcohol
- C) It is specific to hepatocellular injury
- D) It decreases in biliary obstruction

DIFFICULT TIER (ALP, GGT, ALT, AST ratios)

Q15 • Card 75/90

Correct Answer

B

It is induced by alcohol

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q1 • Card 76/90

**Which transporter is defective in
Dubin-Johnson syndrome?**

- A) MRP2 (ABCC2)
- B) OATP1B1
- C) BSEP
- D) MDR3

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q1 • Card 76/90

Correct Answer

A

MRP2 (ABCC2)

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q2 • Card 77/90

What is the molecular defect in Crigler-Najjar syndrome type I?

- A) Complete absence of UGT1A1 activity
- B) Partial deficiency of UGT1A1
- C) Defective MRP2 transporter
- D) Defective OATP1B1

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q2 • Card 77/90

Correct Answer

A

Complete absence of UGT1A1
activity

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q3 • Card 78/90

Which genetic syndrome is associated with a black liver on biopsy?

- A) Gilbert syndrome
- B) Crigler-Najjar syndrome
- C) Dubin-Johnson syndrome
- D) Rotor syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q3 • Card 78/90

Correct Answer

C

Dubin-Johnson syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q4 • Card 79/90

What causes the black pigmentation in Dubin-Johnson syndrome?

- A) Melanin deposition
- B) Bilirubin polymer accumulation
- C) Iron deposition
- D) Copper accumulation

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q4 • Card 79/90

Correct Answer

B

Bilirubin polymer accumulation

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q5 • Card 80/90

In Rotor syndrome, which transporter is defective?

- A) MRP2
- B) OATP1B1/1B3
- C) BSEP
- D) MDR3

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q5 • Card 80/90

Correct Answer

B

OATP1B1/1B3

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q6 • Card 81/90

**Which of the following
differentiates Dubin-Johnson from
Rotor syndrome?**

- A) Urinary coproporphyrin ratio
- B) Serum bilirubin level
- C) ALP elevation
- D) ALT elevation

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q6 • Card 81/90

Correct Answer

A

Urinary coproporphyrin ratio

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q7 • Card 82/90

The MRP2 transporter is responsible for:

- A) Bilirubin uptake into hepatocytes
- B) Bilirubin conjugation
- C) Bilirubin excretion into bile
- D) Bilirubin binding to albumin

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q7 • Card 82/90

Correct Answer

C

Bilirubin excretion into bile

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q8 • Card 83/90

Which condition shows normal urinary coproporphyrin excretion?

- A) Dubin-Johnson syndrome
- B) Rotor syndrome
- C) Gilbert syndrome
- D) Crigler-Najjar syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q8 • Card 83/90

Correct Answer

C

Gilbert syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q9 • Card 84/90

What is the inheritance pattern of Crigler-Najjar syndrome type I?

- A) Autosomal dominant
- B) Autosomal recessive
- C) X-linked
- D) Mitochondrial

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q9 • Card 84/90

Correct Answer

B

Autosomal recessive

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q10 • Card 85/90

**Which gene mutation causes
Dubin-Johnson syndrome?**

- A) UGT1A1
- B) ABCC2
- C) SLCO1B1
- D) ABCB11

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q10 • Card 85/90

Correct Answer

B

ABCC2

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q11 • Card 86/90

A patient with Dubin-Johnson syndrome has which bile pigment in urine?

- A) Bilirubin
- B) Coproporphyrin I
- C) Coproporphyrin III
- D) Urobilinogen

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q11 • Card 86/90

Correct Answer

C

Coproporphyrin III

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q12 • Card 87/90

**What is the typical bilirubin level in
Dubin-Johnson syndrome?**

- A) < 1 mg/dL
- B) 2-5 mg/dL
- C) 10-20 mg/dL
- D) > 30 mg/dL

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q12 • Card 87/90

Correct Answer

B

2-5 mg/dL

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q13 • Card 88/90

Which syndrome is caused by mutations in *SLCO1B1* and *SLCO1B3*?

- A) Dubin-Johnson syndrome
- B) Rotor syndrome
- C) Gilbert syndrome
- D) Crigler-Najjar syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q13 • Card 88/90

Correct Answer

B

Rotor syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q14 • Card 89/90

In Crigler-Najjar type II, what is the response to phenobarbital?

- A) No response
- B) Decrease in bilirubin by $> 30\%$
- C) Increase in bilirubin
- D) Normalization of bilirubin

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q14 • Card 89/90

Correct Answer

B

Decrease in bilirubin by > 30%

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q15 • Card 90/90

Which of the following syndromes involves defective bile salt export?

- A) Dubin-Johnson syndrome
- B) PFIC type 2
- C) Rotor syndrome
- D) Gilbert syndrome

VERY DIFFICULT TIER (MRP2 Transporters,
Genetic Defects)

Q15 • Card 90/90

Correct Answer

B

PFIC type 2